

PROJECT PROPOSAL

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Title:

A Comparative Analysis of Classical Machine Learning and Transformer-Based Models for Sentiment Analysis in Roman Urdu

1. Research Problem and Motivation

Roman Urdu—Urdu text written using the Latin script—is widely used in digital communication across South Asia, especially in countries like Pakistan and India. Its informal nature presents unique challenges for natural language processing (NLP), including:

- Non-standard spellings (e.g., "rasta," "rastey," "rastay" for the same word)
- Limited availability of labeled datasets
- Grammatical flexibility and frequent use of slang

While prior work has explored sentiment analysis in Roman Urdu, most studies focus on binary (positive/negative) classification using classical machine learning methods. The emergence of transformer-based models offers new opportunities for handling linguistic variation and context, but a systematic comparison between classical and modern approaches is lacking.

This research aims to fill that gap by:

- Comparing classical ML models (e.g., SVM, Logistic Regression) with fine-tuned transformer models (e.g., XLM-RoBERTa)
- Evaluating how each approach handles spelling variations and informal language
- Providing a benchmark dataset and performance baseline for future work

2. Research Objectives

- Data Curation & Annotation: Collect and label a Roman Urdu dataset with sentiment labels (Positive, Negative, Neutral). Optionally, include emotional sub-labels (e.g., Joy, Anger) for deeper analysis.
- Model Implementation:

- Classical ML Pipeline: Implement TF-IDF (word and character n-grams) with classifiers like SVM and Logistic Regression.
- Transformer Pipeline: Fine-tune multilingual transformer models such as XLM-RoBERTa and DistilBERT on the curated dataset.
- Evaluation & Comparison: Benchmark models using metrics such as accuracy, precision, recall, and macro F1-score. Analyze strengths and weaknesses of each approach.

3. Methodology

Feature Extraction:

- Classical ML: TF-IDF vectors (unigrams, bigrams) and character n-grams
- Transformers: Use tokenizers and embeddings from pre-trained models

Models:

- Classical: SVM, Logistic Regression, Naïve Bayes
- Modern: Fine-tuned XLM-RoBERTa, DistilBERT

Hyperparameter Tuning:

- Use grid search for classical models
- Explore learning rates, number of epochs, and batch sizes for transformers

4. Datasets and Tools

Datasets:

- Primary dataset: UCI Roman Urdu Sentiment Analysis Dataset
- Supplementary data from Hugging Face and Kaggle

Tools & Libraries:

- Python, PyTorch, Transformers (Hugging Face)
- Scikit-learn, NLTK, Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Optional UI: Streamlit for demo

6. Expected Challenges

- Data Scarcity: Use data augmentation (e.g., synonym replacement, back-translation)
- Spelling Variations: Leverage character-level models and transformer tokenization
- Overfitting: Apply regularization and cross-validation

DRAFT

Project Title:

JazbaatAI: Sentiment Analysis for Roman Urdu

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1. Research Problem and Motivation

Research Problem:

In Pakistan and South Asia, Roman Urdu (Urdu written in the English script) dominates digital communication, particularly on **social media platforms** like Twitter, Facebook, and YouTube. These texts often include heavy **code-switching** between Roman Urdu, Urdu (script-based), and English. The challenges include:

- Lack of standardized spelling and grammar in Roman Urdu.
- Complex **code-mixing** within a single sentence.

- Limited **annotated datasets** and resources for Roman Urdu NLP.

Existing NLP systems, mostly trained on English or high-resource languages, fail to interpret **semantics, sentiment, and intent** in this noisy, low-resource setting.

Motivation:

1. **Low-Resource Language Challenge:** Roman Urdu represents a **low-resource and underrepresented language** in NLP, despite being used daily by millions.
2. **Real-World Impact:**
 - **Social Media Monitoring:** Track public opinion, detect misinformation, and identify trends.
 - **Mental Health Applications:** Spot negative emotions or distress signals in user-generated content.
 - **Accessibility & Inclusivity:** Enable better understanding for systems trained mainly on English or formal Urdu.
3. **Research Contribution:** Develop novel techniques for handling **code-switching, transliteration, and sentiment analysis** in noisy, multilingual data. Insights can extend to other underrepresented languages globally.

2. NLP Techniques to Be Used

We propose a **two-module system**, with an **incremental roadmap**:

Module 1: Translation Pipeline (Roman Urdu → Urdu → English)

- **Step 1: Text Normalization & Language Identification**
 - Token-level tagging for Urdu/English.
 - Rule-based + heuristic cleaning.
- **Step 2: Transliteration (Roman Urdu → Urdu Script)**
 - Fine-tuned **Seq2Seq model** (Transformer-based).

- Alternatively, adapt pre-trained transliteration tools (e.g., Google's Transformer Transliteration).
- **Step 3: Machine Translation (Urdu → English)**
 - Fine-tune multilingual NMT models like **mBART**, **mT5**, **M2M-100** on available Urdu-English parallel corpora.

Module 2: Emotion & Sentiment Analysis

- **Direct Emotion Classification (Baseline):**
 - Fine-tune **XLM-RoBERTa**, **mBERT**, or **IndicBERT** on Roman Urdu sentiment data.
 - Input: Original Roman Urdu text → Output: Emotion label.
 - **Hybrid Approach (Research Focus):**
 - Feed translated English text into a second sentiment classifier.
 - Combine predictions from both models (weighted ensemble).
 - Goal: Improve robustness by reducing errors from one pipeline alone.
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3. Datasets and Tools to Be Used

Datasets

- **Transliteration & Translation:**
 - **Roman-Urdu ↔ Urdu parallel corpus:** Scraped from Twitter, YouTube, blogs (to be cleaned & normalized).
 - **Urdu-English parallel corpora:** Common Crawl, OPUS, or CCMatrix.
- **Emotion & Sentiment Analysis:**

- **Primary Dataset:** Manually annotated Roman Urdu sentences with emotions (Happy, Sad, Angry, Neutral).
- **Secondary Datasets:** Urdu sentiment datasets (GitHub, FIRE 2020 Code-Mixed shared task). These will be transliterated into Roman script for consistency.

Tools & Frameworks

Priority 1 (Essential):

- **Frameworks:** PyTorch, Hugging Face Transformers & Datasets.
- **Models:** XLM-RoBERTa, mBERT, DistilBERT, mBART, M2M-100.
- **Preprocessing:** urduhack, deep-transliteration, regex, langdetect.
- **Data Handling:** Pandas, NumPy.

Priority 2 (Highly Useful):

- **Sentence Transformers:** For embeddings & similarity checks.
- **APIs (OpenAI, LLaMA):** For zero-shot baselines or data augmentation.
- **Evaluation Metric:** BLEU for translation, F1 for sentiment.

Priority 3 (Advanced / Stretch Goals):

- **LangChain / LangGraph:** For chaining translation + classification.
- **Model Deployment:** Gradio / Streamlit (demo UI), FastAPI (backend).
- **Advanced Models:** LLaMA fine-tuning (if resources allow).

4. Evaluation Metrics

The system will be evaluated at **module-level** and **end-to-end**:

- **Module 1 (Translation Pipeline):**
 - Transliteration Quality: **Word Error Rate (WER)**, **BLEU score**.
 - Translation Quality: **BLEU**, **TER (Translation Edit Rate)**, **human evaluation** for fluency.
 - **Module 2 (Emotion Analysis):**
 - **Accuracy, Precision, Recall, F1-score (macro-averaged)** → F1 is primary due to class imbalance.
 - **End-to-End System:**
 - Human Evaluation: Judges will assess whether the translation + sentiment classification make sense together.
 - **Ablation Studies:** Compare (1) direct sentiment model, (2) translation-based sentiment model, (3) hybrid ensemble.
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5. Project Roadmap (Phased Implementation)

Phase 1 – Baseline Setup (Weeks 1–2):

- Collect small code-mixed dataset.
- Fine-tune **XLM-RoBERTa** for direct sentiment classification.

Phase 2 – Transliteration + Translation (Weeks 3–4):

- Build Roman Urdu → Urdu transliteration model.
- Fine-tune MT model (Urdu → English).

Phase 3 – Hybrid Sentiment Analysis (Weeks 5–6):

- Combine outputs of direct model + translated pipeline.
- Perform ensemble experiments.

Phase 4 – Evaluation & Reporting (Weeks 7–8):

- Run module-level + end-to-end evaluations.
 - Human evaluation + ablation study.
 - Write final report + build demo (if time).
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6. Expected Contributions

- Creation of a **novel Roman Urdu sentiment dataset**.
- Experimental comparison of **direct vs. hybrid sentiment analysis approaches**.
- Insights into **low-resource, code-mixed NLP challenges** with global relevance.
- (Stretch Goal) End-to-end system: Roman Urdu → English translation + sentiment classification.